

Static hard-sphere scattering with fixed molecular coordinates

CPU implementation · isotropic incidence · native BVH nearest-hit queries · specular reflection

a Prepare geometry and incident rays

Fixed atomic coordinates + effective radii
Spatial median groups → padded AABB hierarchy

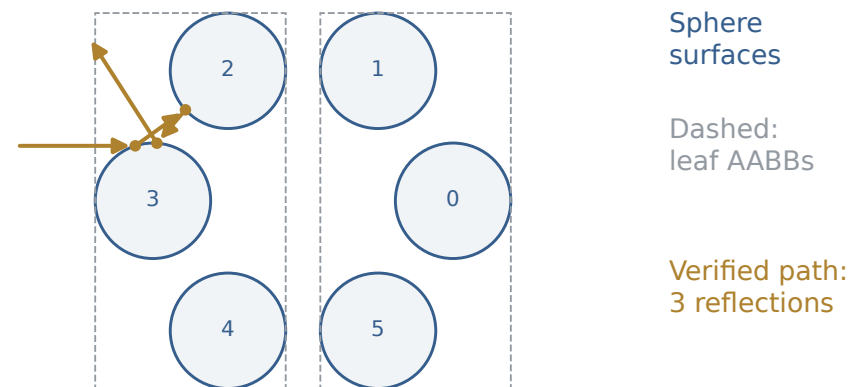


Four-dimensional scrambled Sobol samples
Direction: uniform on S^2
Impact position: uniform on a perpendicular disk

$$R = \max_i (\|\mathbf{c}_i - \bar{\mathbf{c}}\| + r_i) (1 + 10^{-9})$$

Three independent blocks; total incident count $N = 3n$.

b Follow the nearest physical collision



Synthetic planar section; illustrative ray only.

c Reflect, escape, or retain a residual bound

BVH traversal → exact nearest sphere entry



At a hit, use the physical surface normal $\hat{\mathbf{n}}$

$$\mathbf{v} = \mathbf{u} - 2(\mathbf{u} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$$

Repeat the nearest-hit query with the reflected direction.

No further hit → escaped contribution

Cap reached with another hit → unresolved; retain its bound

d Estimate CCS and report the measured cost

$$\hat{\Omega} = \frac{\pi R^2}{N} \sum_{j \in \text{escaped}} (1 - \mathbf{u}_{j, \text{in}} \cdot \mathbf{u}_{j, \text{out}})$$

$$T_{\text{sample}} = 2 \frac{\pi R^2}{N} N_{\text{unresolved}}$$

Bound applies only to unresolved sampled paths.

Warm components
Prepare geometry + source + trace + readout

Separate process wall: start → import/cache/input → output → exit

Incident disk positions and directions vary; the molecular coordinates remain fixed. The depicted path is not a performance measurement.